TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure diagram of a topological insulator shown in the diagram below, which features band inversion between the bulk valence and conduction bands. Explain how this inversion leads to a non-trivial topology.

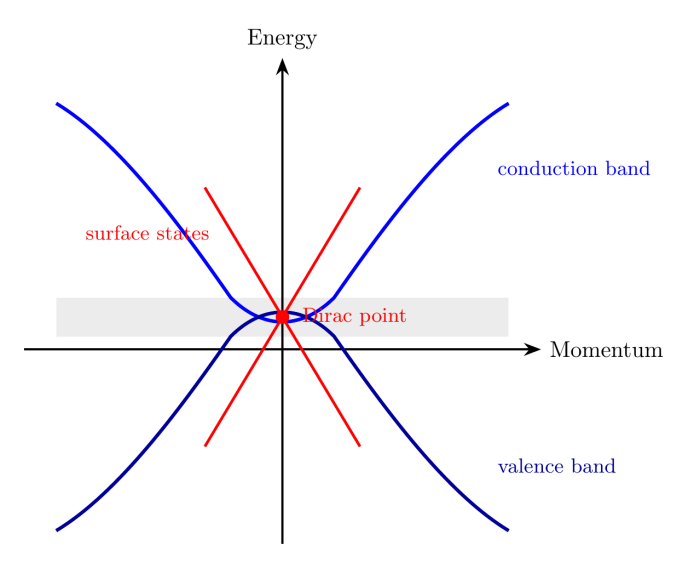

Fig. 1. A band structure diagram of a topological insulator. The diagram should show the bulk valence band and conduction band with band inversion. The valence band should be depicted as crossing over the conduction band, indicating the inversion. Include labels for the valence band, conduction band, and the Dirac point where the bands touch. Highlight the surface states that cross the band gap, indicating the non-trivial topology. Use a V-shaped linear Dirac cone to represent the surface states crossing through the gap.

Question 2 10.0 points

Interpret the bulk-boundary correspondence as illustrated in the surface state diagram of a topological insulator in the diagram below. Describe how the surface states are protected by topology.

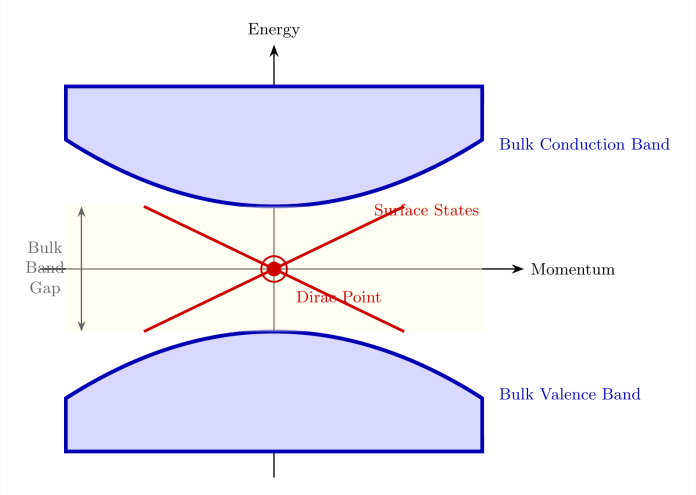

Fig. 2. A surface state diagram of a topological insulator. The diagram should show the bulk bands as parabolic curves with a gap, and the surface states as linear Dirac cones crossing through the gap. The bulk bands should be labeled as 'Bulk Valence Band' and 'Bulk Conduction Band', and the surface states should be labeled as 'Surface States'. The Dirac point where the surface states cross should be highlighted to indicate the topological protection. The axes should be labeled with 'Energy' on the vertical axis and 'Momentum' on the horizontal axis.

Question 3

10.0 points

Examine the edge state diagram shown below associated with the quantum spin Hall effect. Explain how these states allow for dissipationless edge transport.

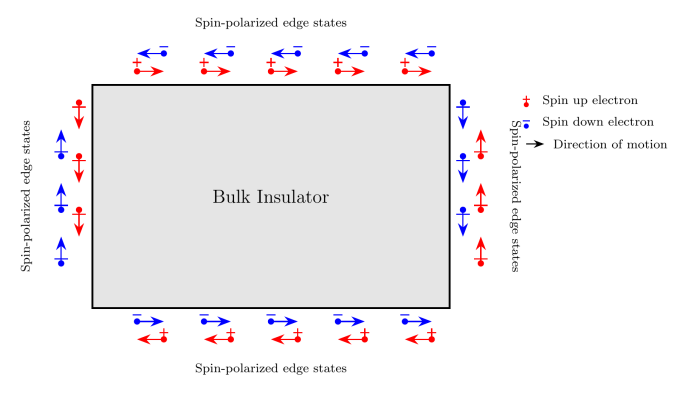

Fig. 3. A diagram illustrating the edge state configuration in the quantum spin Hall effect. The diagram should show a two-dimensional topological insulator with edge states along the boundaries. Use arrows to indicate the direction of electron spin and movement along the edges, highlighting the helical nature of the edge states. Label the bulk insulator region and the conducting edge states. Emphasize the dissipationless transport along the edges due to the spin-momentum locking characteristic of these states. IMPORTANT correction needed: The diagram contains a legend entry that explicitly states the core physical principle the student is asked to explain, constituting an answer leak.. Specifically: The legend text "Opposite spins propagate in opposite directions (spin-momentum locking)" directly provides the key information needed to answer the question "Explain how these states. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 4 10.0 points

Analyze the k-space structure of a 2D system with a Z2 topological invariant as depicted in the diagram below. Explain the significance of the Z2 invariant.

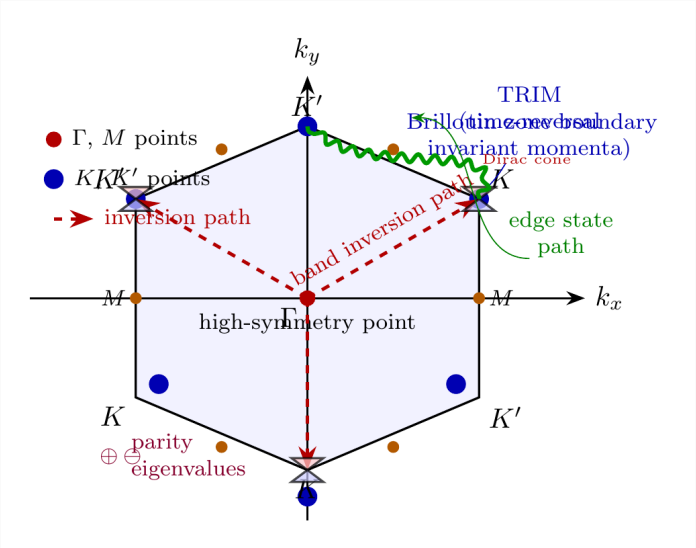

Fig. 4. A 2D k-space structure diagram for a system with a Z2 topological invariant. The diagram should include a depiction of the Brillouin zone, with Dirac cones representing the band structure. Highlight the points of band inversion and indicate the presence of edge states. The Z2 invariant should be represented by a label or symbol indicating its topological nature. Use clear labels for the k-space axes and any significant features.

Question 5 10.0 points

Analyze the Dirac cone dispersion relation depicted in the diagram below on the surface of a 3D topological insulator. Discuss how this structure is related to massless Dirac fermions.

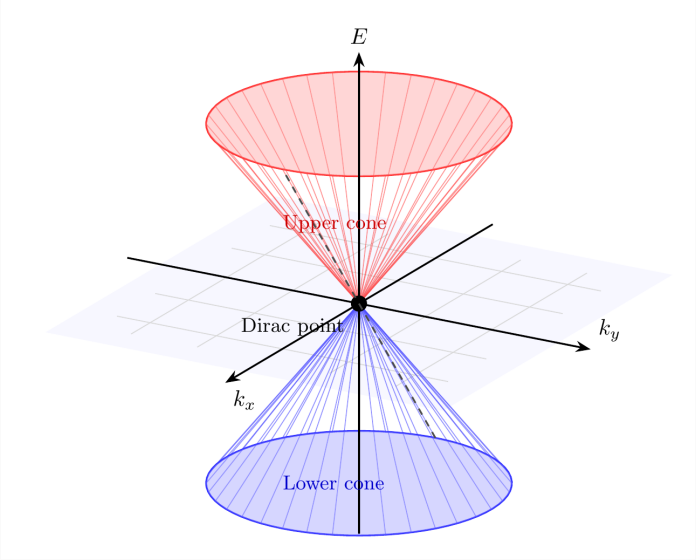

Fig. 5. Dirac cone dispersion relation on the surface of a 3D topological insulator. The diagram should show a 3D plot with two cones touching at a single point (the Dirac point) at the origin. The cones should be symmetric about the vertical axis, representing energy (E), with the horizontal axes representing momentum (k_x and k_y). Label the axes as E, k_x, and k_y. The cones should be linear, illustrating the linear energy-momentum relationship characteristic of massless Dirac fermions.